

National University of Computer & Emerging Sciences
FAST-Karachi Campus
CS4051- Information Retrieval
Quiz#2 (Double Weightage)

Dated: April 29, 2026

Marks: 20

Time: 20 min.

Std-ID: _____ Sol _____

Question No. 1

Match the two columns appropriately. [5]

Column A	Column B
1. Multinomial Naive Bayes (C)	A. Classifier that assigns a document to the majority class among its nearest training documents
2. Bernoulli Naive Bayes (F)	B. Classification model based on real-valued document vectors such as tf-idf vectors
3. Laplace smoothing (E)	C. Uses term frequency and considers multiple occurrences of words
4. Rocchio classifier (D)	D. Learns one centroid or prototype vector for each class
5. kNN classifier (A)	E. Adds one to counts to avoid zero probabilities
6. Feature selection (H)	F. Uses binary presence or absence of terms
7. Contiguity hypothesis (G)	G. Documents of the same class occupy a continuous region in vector space
8. Linear classifier (I)	H. Selects a subset of useful terms and removes noisy features
9. Vector space classification (B)	I. Classifier whose decision boundary is a hyperplane
10. Bias–variance tradeoff (J)	J. Explains why a more powerful model may over fit noisy training data

Question No. 2

Suppose the rare word “arachnocentric” occurs only in documents labeled Sports in the training set, but it has no real relationship with sports. What sort of problem this can cause? Suggest at least 2 approaches to resolve this problem.

This word causes overfitting because the classifier learns an accidental training-set pattern as if it were a real class signal. The best solution is feature selection, especially removing rare/noisy terms and keeping only terms that consistently provide useful information for classification.

Question No.3

Consider the following examples for the task of text classification [5+5]

	docID	words in document	in $c = \text{China?}$
training set	1	Taipei Taiwan	yes
	2	Macao Taiwan Shanghai	yes
	3	Japan Sapporo	no
	4	Sapporo Osaka Taiwan	no
test set	5	Taiwan Taiwan Sapporo	?

- a. Using the k-Nearest Neighbors (KNN) with $k=3$ identify the class of test instance docID=5?

Consider the features

$V = \langle \text{japan; macao; osaka; sapporo; shanghai; taipei; taiwan} \rangle$

$D1 = \langle 0,0,0,0,0,1,1 \rangle$ magnitude $|D1| = \sqrt{2}$ $D2 = \langle 0,1,0,0,1,0,1 \rangle$ magnitude $|D2| = \sqrt{3}$
 $D3 = \langle 1,0,0,1,0,0,0 \rangle$ magnitude $|D3| = \sqrt{2}$ $D4 = \langle 0,0,1,1,0,0,1 \rangle$ magnitude $|D4| = \sqrt{3}$
 $D5 = \langle 0,0,0,1,0,0,2 \rangle$ magnitude $|D5| = \sqrt{5}$

Now distance as (dot-product) with D5 for each document can be given as

Dot-Product (D1,D5) = 2 Dot-Product (D2,D5) = 2
 Dot-Product (D3,D5) = 1 Dot-Product (D4,D5) = 3

For $K=3$ the three nearest neighbors of D5 will be D1, D2 and D3 from these D1 and D2 majority belong to class $c=\text{china}$ so the DocID=5 will be $c=\text{yes}$

- b. Using the Rocchio's algorithm, classify the test instance docID=5?

Consider the two mass vectors of the classes

$$\begin{aligned} \mu(\text{class}=\text{china}=\text{Yes}) &= \frac{1}{2} (D1+D2) = \frac{1}{2} \{ \langle 0,0,0,0,0,1,1 \rangle + \langle 0,1,0,0,1,0,1 \rangle \} \\ &= \langle 0,1/2,0,0,1/2,1/2,1 \rangle \end{aligned}$$

$$\begin{aligned} \mu(\text{class}=\text{china}=\text{No}) &= \frac{1}{2} (D3+D4) = \frac{1}{2} \{ \langle 1,0,0,1,0,0,0 \rangle + \langle 0,0,1,1,0,0,1 \rangle \} \\ &= \langle 1/2,0,1/2, 1,0,0,1/2 \rangle \end{aligned}$$

Now angle between vectors from $\mu(\text{class}=\text{china}=\text{Yes})$ to D5 can be computed as using:

$$\begin{aligned} \text{angle}(\mu(\text{class}=\text{china}=\text{Yes}), D5) &= 0.8283 \\ \text{angle}(\mu(\text{class}=\text{china}=\text{No}), D5) &= 0.8283 \end{aligned}$$

As the D5 has the same angle from the two center mass Rocchio's cannot be able to decide about this example.